SAFETY SIGNAL REVIEW CAPSTONE

Aggregate adverse-reaction dataset review | Milestone 5 of 5

Purpose. Summarize the available case counts, seriousness, outcomes and reporting patterns; identify a potential safety concern; consider alternative explanations; and recommend follow-up evidence and risk-minimization questions.

Important limitation: This file contains only eight reports, with a different suspected drug and reaction in every row. These spontaneous/sample reports do not establish incidence, comparative risk, or causality. The potential signal below is a priority hypothesis for investigation, not a confirmed signal.

1. Executive summary

The dataset contains **8 reports** involving eight different suspected medicines and eight distinct reported reactions. Three reports (3/8, 37.5%) have the outcome recorded as hospitalised; two reports are marked severe and one life-threatening. Three patients are recorded as recovered, one as recovering, one as not recovered, and three as hospitalised (the outcome field does not separately indicate final recovery for these cases). The most urgent individual case for follow-up is ADR-9007 (warfarin; uncontrolled hemorrhage), marked life-threatening and hospitalised.

2. Dataset overview

Measure	Finding
Total reports	8
Age	Range 28–71 years; mean 52.0 years
Gender	4 female; 4 male
Reaction onset	1–30 days; mean 11.2; median 8.5 days
Severity	3 mild; 2 moderate; 2 severe; 1 life-threatening
Outcome	3 recovered; 1 recovering; 3 hospitalised; 1 not recovered
Reporter type	Physician: 3; Pharmacist: 2; Patient: 2; Oncologist: 1

3. Case-level summary

Report ID	Suspected drug	Reported reaction	Onset (days)	Severity	Outcome
ADR-9001	Amoxicillin	Skin Rash & Hives	2	Mild	Recovered
ADR-9002	Atorvastatin	Severe Muscle Myalgia	14	Moderate	Recovering
ADR-9003	Ciprofloxacin	Tendonitis	7	Severe	Hospitalized
ADR-9004	Ibuprofen	Gastric Ulcer Bleed	5	Severe	Hospitalized
ADR-9005	Lisinopril	Persistent Dry Cough	21	Mild	Recovered
ADR-9006	Metformin	Nausea & Diarrhea	1	Mild	Recovered
ADR-9007	Warfarin	Uncontrolled Hemorrhage	10	Life-Threatening	Hospitalized
ADR-9008	Paclitaxel	Peripheral Neuropathy	30	Moderate	Not Recovered

4. Reporting patterns and interpretation

Seriousness/severity: 3 of 8 reports (37.5%) are labelled severe or life-threatening. Three reports (37.5%) have hospitalisation recorded as the outcome. Severity labels and regulatory seriousness are not interchangeable; the case narratives and hospitalization details should be verified.

Outcomes: 3 reports (37.5%) are recorded as recovered. One case is recovering and one is not recovered. Three records say hospitalised; this may describe disposition rather than final outcome, so follow-up is needed to clarify final status.

Timing: Reported onset ranges from 1 to 30 days (median 8.5). The reactions and medicines differ, so the timing range alone does not establish a shared pattern.

Reporting trend: The file has no report dates, exposure denominators, or repeated drug–event pairs. A time trend, disproportionality statistic, or incidence estimate cannot be calculated from this dataset.

5. Potential safety signal / priority hypothesis

Priority case hypothesis: Warfarin-associated uncontrolled hemorrhage (ADR-9007) merits urgent individual case review because the report is labelled life-threatening and hospitalised. This is a clinically important safety concern, but it is only one report and is not evidence of a newly emerging or statistically supported signal by itself.

Evidence to check: confirm bleeding site and clinical diagnosis; INR/coagulation results and trend; warfarin dose and recent dose changes; indication and treatment dates; adherence; interacting medicines and supplements; comorbidities; trauma or other causes; treatment given; admission/discharge dates; and final outcome. Compare the event with the current approved product information and relevant clinical literature.

The dataset also contains hospitalised tendonitis after ciprofloxacin and gastric ulcer bleeding after ibuprofen. These should be reviewed against the relevant product safety information and clinical circumstances, but the sample is too small to rank product-level risk or infer causation.

6. Alternative explanations and limitations

Possible explanations include underlying disease, concomitant medicines, dose or administration factors, drug interactions, patient-specific risk factors, reporting/stimulated-reporting bias, incomplete or inaccurate coding, and chance. The dataset does not include clinical narratives, medical history, concomitant medication details, dechallenge/rechallenge, report dates, duplicate status, or exposure denominators. These gaps prevent causal conclusions and robust statistical signal detection.

7. Follow-up evidence and risk-minimization questions

Follow-up actions

1. Prioritise ADR-9007 and the other hospitalised cases for complete case narratives and verified outcomes.
2. Confirm dose, route, indication, dates, relevant medical history, concomitant medicines, investigations and treatment.
3. Check duplicates and coding consistency; obtain report dates for temporal analysis.
4. Compare each event with the applicable product label/reference safety information and authoritative clinical evidence.
5. If a larger dataset becomes available, examine repeated drug–event combinations and appropriate exposure/background data; use validated signal-detection methods where suitable.

Risk-minimization questions

- Is the event known and adequately described in the applicable product information?
- Are preventable factors, monitoring gaps, interactions, or dosing issues present?
- Would targeted counselling, monitoring, prescriber communication, or a label review be proportionate to the evidence?
- What additional evidence is needed before any formal benefit–risk or regulatory decision?

8. Conclusion

This eight-report sample contains three hospitalised outcomes and one life-threatening report. The warfarin–hemorrhage case is the highest-priority individual case for follow-up; ciprofloxacin-associated tendonitis and ibuprofen-associated gastric ulcer bleeding also warrant review. Because every drug–reaction combination appears only once and there are no report dates or exposure denominators, no recurring aggregate signal or reporting trend can be confirmed. Further case validation and reference-information review are required before drawing causal or regulatory conclusions.

Training sample only. Not for clinical decision-making or use as a real-patient safety assessment or regulatory submission.